

Exercises, sheet 2. Fix a first-order language L . M and N will denote L -structures and T will denote an L -theory. If $(I, <)$ is a linear order, by a (an elementary) chain of L -structures we mean a collection $(M_i)_{i \in I}$ such that $M_i \subseteq M_j$ if $i < j$ (resp. $M_i \prec M_j$).

1. Show that any for any finite structure in a finite language there is a sentence that determines it up to an isomorphism.
2. Show that isomorphic structures satisfy the same sentences. Hint: induction based on the complexity of the formula.
3. Show for an elementary chain $(M_i)_{i \in I}$, each M_i is elementarily embedded in the union of all M_j 's (define a natural L -structure on this union). Generalise to the case where $(I, <)$ is a directed system.
4. For a theory T let $T_\forall$ be the set of universal sentences that are consequences of T . Show that $M \models T_\forall$ if and only if there is an $N \models T$ with $M \subseteq N$.
5. Deduce that a formula $\varphi(\bar{x})$ is T -equivalent to a universal formula if and only if for every $N \subseteq M$, $M, N \models T$ and $\bar{a} \in N^{|\bar{x}|}$, we have

$$M \models \varphi(\bar{a}) \implies N \models \varphi(\bar{a}).$$

6. Show that $\text{Diag}_{\text{el}}(M)$ (resp. $\text{Diag}(M)$) is consistent with T , then there is an elementary embedding (resp. an embedding of L -structures) of M in a model of T .
7. We call T *model complete* if for every model $M \models T$, the $L(M)$ -theory $T \cup \text{Diag}(M)$ is complete (where $L(M) = L \cup \{c_m : m \in M\}$). Show that the following are equivalent:
 - T is model complete,
 - every embedding of models of T is elementary,
 - every formula is T -equivalent to a universal one,
 - every formula is T -equivalent to an existential one.
8. Show that a theory T can be axiomatized by $\forall\exists$ -formulas (i.e., T is *inductive*) if and only if for any chain of L -structures $(M_i)_{i \in I}$ that are models of T , their union is also a model of T .
9. A model $M \models T$ is called *existentially closed* if for any extension $M \subseteq N$ with $N \models T$, M is *existentially closed in* N (written $M \prec_\exists N$), i.e., whenever $\varphi(\bar{y})$ is an existential formula and $\bar{b} \in M^{|\bar{y}|}$, then

$$M \models \varphi(\bar{b}) \iff N \models \varphi(\bar{b}).$$

Show that if T is inductive, then any model of T can be embedded in an existentially closed model of T . Show that T is model complete if and only if all of its models are existentially closed.

10. An L -theory T^* is a *model companion* of T if:
 - every model of T admits an embedding into a model of T^* ,
 - every model of T^* admits an embedding into a model of T ,
 - T^* is model complete.

Show that if a model companion of T exists, it is unique up to equivalence of theories.

11. Show that if T is inductive, its model companion exists if and only if the class of its existentially closed models is axiomatized by a theory (i.e., this class is *elementary*).